

Deep Learning Pipeline Workflow

STANDARD OPERATING PROCEDURE & EXECUTION FRAMEWORK

PART I: DATA UNDERSTANDING, EXPLORATION & PREPARATION

Data Collection & Sampling

- Count numerical, categorical, image, text, or audio features.
- Inspect missing data, corrupt samples, and check data integrity.
- Drop unnecessary, redundant, or non-informative features.

Data Information & Structure

- Execute metadata checks, data types, tensor shapes, and dimensions.

Statistical & Distribution Analysis

- Analyze percentile values, skewness, and class distribution.
- Check standard deviation, mean, and value ranges across features.

Exploratory Data Analysis (EDA) & Visualization

- Analyze graphs for features relative to target variables.
- Perform dimensionality visualization (e.g., t-SNE, UMAP, PCA) for high-dimensional data.

PART II: PREPROCESSING & DATA PIPELINE BUILDING

Outlier Detection & Noise Filtering

- Identify anomalous data, apply clipping, or remove noisy samples.

Column Transformation & Augmentation

- **Scaling:** Standardize/Normalize continuous variables or pixel values (e.g., [0, 1] range).
- **Encoding:** Apply One-Hot Encoding, Ordinal Encoding, or Tokenization/Embeddings for text/categorical features.
- **Data Augmentation:** Apply transformations (image rotation, audio pitch shift, text back-translation) to expand dataset diversity.
- **Imputation:** Handle missing values using standard techniques or DL-based imputation.

Data Loading & Batching Pipeline

- Split dataset into Train, Validation, and Test sets.
- Set up data loaders (e.g., PyTorch DataLoader or tf.data.Dataset) with batching, shuffling, and prefetching.

PART III: MODEL ARCHITECTURE & TRAINING

Model Architecture Selection

- Choose target neural network structure (e.g., CNN, RNN, Transformer, Autoencoder, MLP).
- Consider pre-trained models for Transfer Learning.

Loss Function & Optimizer Setup

- Define task-specific loss function (e.g., Cross-Entropy, MSE, Focal Loss).
- Select optimizer (e.g., Adam, AdamW, SGD) and configure learning rate schedulers.

Model Training & Regularization

- Apply regularization: Dropout, Batch Normalization, Weight Decay, and Early Stopping.
- Execute model training loop across epochs.

Hyperparameter Tuning

- Perform automated hyperparameter optimization (e.g., Learning Rate, Batch Size, Layer Count) using Optuna or Ray Tune.

PART IV: EVALUATION, VALIDATION & OPTIMIZATION

Model Evaluation & Error Analysis

- Monitor Loss/Accuracy curves for underfitting and overfitting. If issues exist, loop back to steps in Part I-III.

Evaluation Metrics

- **Classification:** Confusion Matrix, Accuracy, Precision, Recall, F1-Score, ROC-AUC.
- **Regression:** MAE, MSE, RMSE, R-squared.

Cross-Validation & Generalization

- Validate model performance across K-Folds or out-of-distribution test sets.

PART V: OPTIMIZATION & DEPLOYMENT

Model Export & Compression

- Export model to production format (e.g., ONNX, TensorRT, TFLite) with quantization/pruning.

FINAL VERDICT & SIGN-OFF

Signed off by: _____

Date: _____